



The Question / Hypothesis Investigated

Our team investigated the robustness of a Principal Component Analysis (PCA) facial recognition model. Specifically, we hypothesized and tested how much the model's recognition accuracy degrades when test images are subjected to various real-world environmental noises (Gaussian, Salt & Pepper, Poisson) and partial occlusions (fixed geometric masks and random sliding blocks).

The Most Important Conclusion Reached

The most critical finding is that standard PCA is fundamentally a "global" algorithm and fails catastrophically against targeted occlusion. Because PCA looks at the entire pixel grid as one interconnected vector, it cannot ignore missing data. Instead, it processes the black pixels of a mask as actual facial structure. This corrupts the projection weights, forcing the model to confidently hallucinate bizarre and incorrect facial geometry during reconstruction; an effect that actually worsens as the number of principal components (k) increases.

Examples of Success and Failure

- **Example of Success (Gaussian Noise Resilience):**
While highly vulnerable to most extreme distortions, the model demonstrated slight resilience against Gaussian noise at maximum intensity, retaining ~20% accuracy. Because Gaussian noise applies a symmetrical bell curve of random values, the overarching facial geometry survives just enough for the Nearest Neighbor algorithm to occasionally find the correct match.
- **Example of Failure (Poisson & Salt & Pepper Destruction):**
The model proved virtually useless against extreme Poisson and Salt & Pepper noise. Salt & Pepper flatlined accuracy to near 0% because the extreme black and white pixels completely blinded the eigenvectors and destroyed local geometric structures. Similarly, at peak intensity, Poisson noise introduced massive, chaotic brightness spikes that totally overwhelmed the model, dropping accuracy to roughly 10%.